

Lecture 6

Measures of Spread

Review

1 One huge outlier is added to a dataset. Which moves more, the mean or the median?

The mean is pulled toward it. The median barely moves – it only cares about position in the ordered list.

2 Besides the sum over n , what were the other two ways we wrote the mean?

From a frequency table, and from a relative frequency table. All three give the same number.

3 What does it mean to say a score sits at the 75th percentile?

75 percent of the observations fall at or below that score.

4 You already have the median. How do you get Q1 and Q3 from it?

Q1 is the median of the lower half, Q3 the median of the upper half – leaving the median itself out of both halves when n is odd.

Review: quartiles of the exam scores

- Last time you were asked for Q1, Q2 and Q3 of these 15 exam scores
 - 61 61 65 65 66 68 69 73 74 75 76 78 79 90 94
- n is odd, so start at the median and work outward
- Keep these three numbers - every example today uses them

Working:

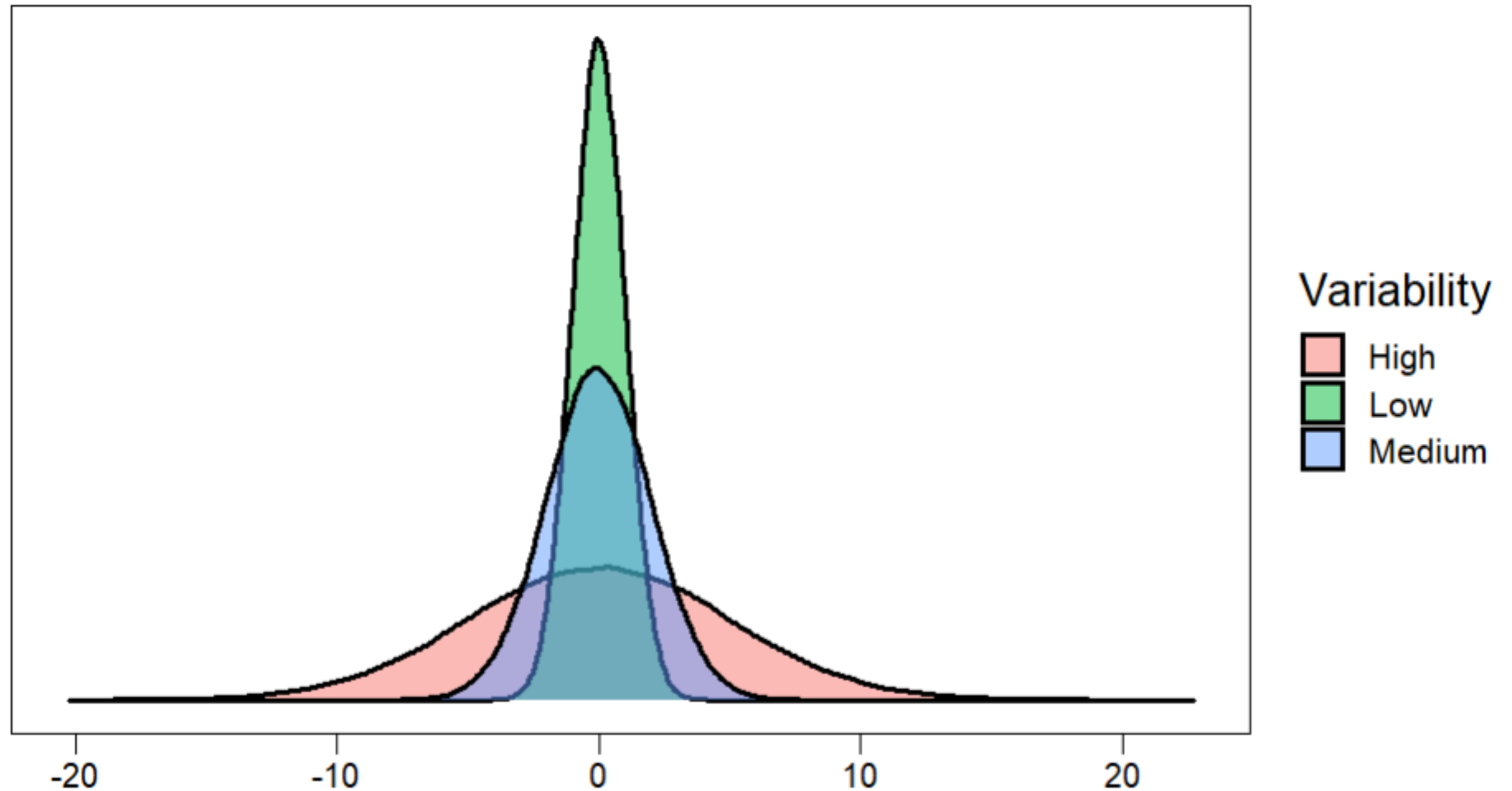
Q2 is the median: with $n = 15$ that is the 8th value $\rightarrow Q2 = 73$

lower half 61 61 65 65 66 68 69
its median $\rightarrow Q1 = 65$

upper half 74 75 76 78 79 90 94
its median $\rightarrow Q3 = 78$

The median itself belongs to neither half.

Variability of A Distribution: Measures of Spread

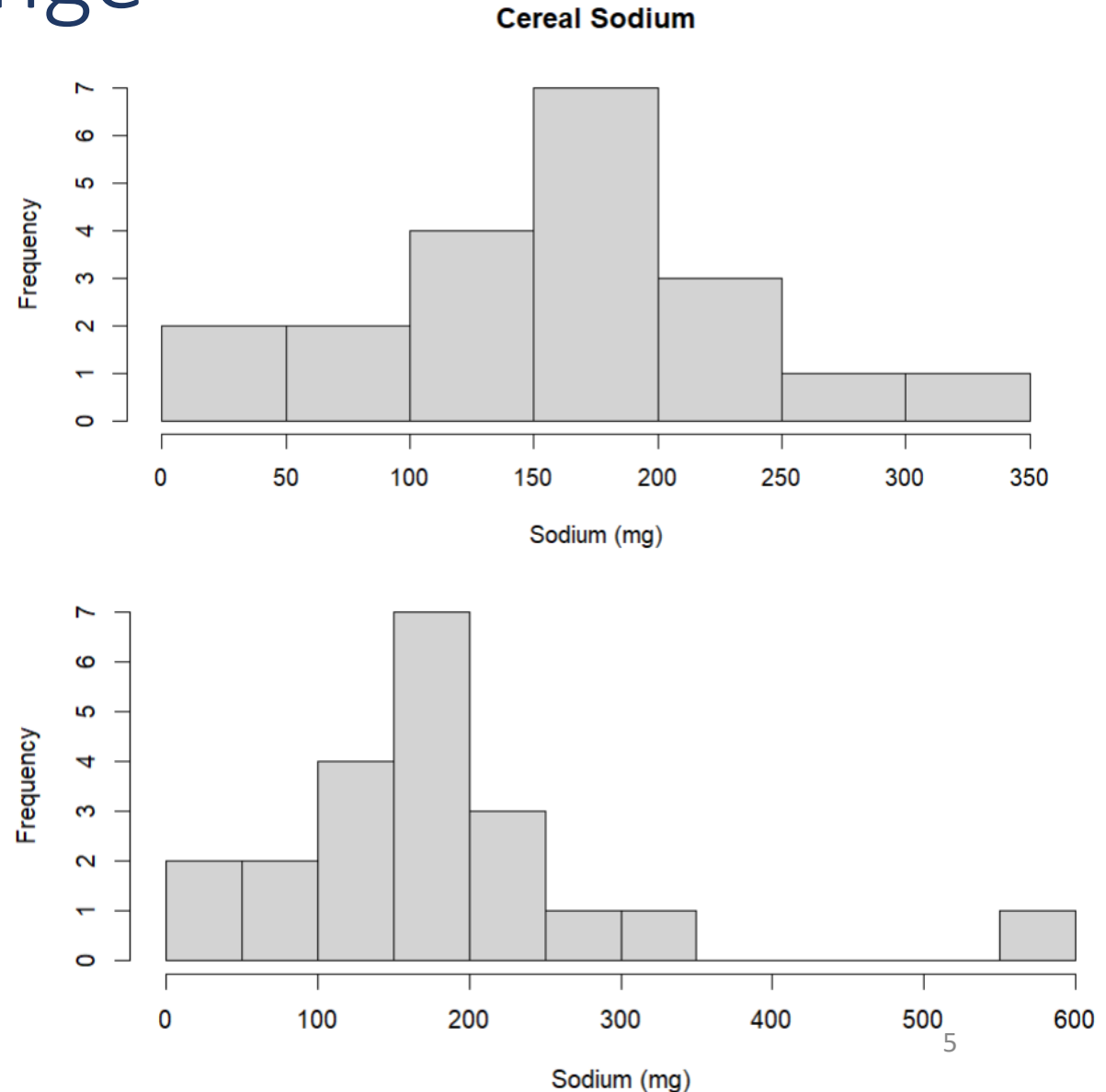


Measures of Spread: Range

- The **range** is a measure of the distance between the smallest and largest values in the data

The range can be computed with only two data points the minimum value and maximum value

- If the range of a set of data is large, then usually this indicates greater dispersion of values
- The range is severely affected by the presence of outliers
- We typically do not use the range to measure variability



Measures of Spread: Interquartile Range

- The **interquartile range (IQR)** measures the spread of the middle 50% of the observations
- It is resistant to outliers

$$IQR = Q3 - Q1$$

- The more variability the larger the value of the **IQR**
- **IQR** is a good choice for distributions that are highly skewed!

Practice: Finding quartiles and IQR

Exam Scores 61,61,65,65,66,68,69,73,74,75,76,78,79,90,94

Compute the IQR

Working:

$Q1 = 65$ $Q3 = 78$ (from the review slide)

$IQR = Q3 - Q1 = 78 - 65 = 13$

The middle half of the class scored within 13 points of one another.

Measures of Spread: Deviation

- A better measure of variability that uses *all* the data is based on **deviations**
- **deviations** are the distances of each value from the mean of the data:

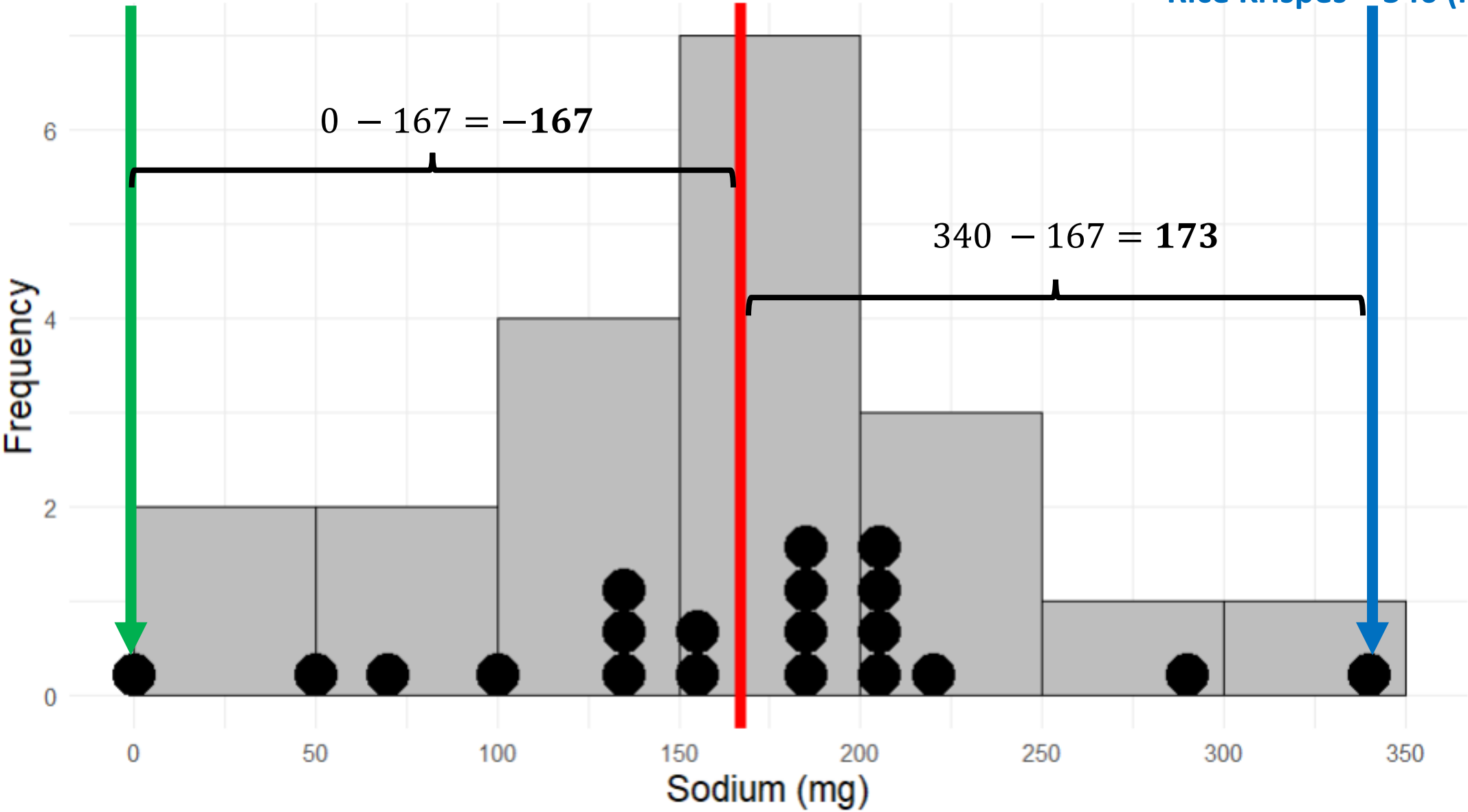
Deviation of an observation $x_i = (x_i - \bar{x})$

- Every observation will have a deviation from the mean

Frosted Mini Wheats = 0 (mg)

Mean = 167 (mg)

Rice Krispes = 340 (mg)



Measures of Spread: Variance

- The sum of all deviations is zero. $\sum_{i=1}^n (x_i - \bar{x}) = 0$
- We typically use either the **squared deviations** or their **absolute value**
Squared deviation of an observation $x_i = (x_i - \bar{x})^2$
- The **Variance** of a distribution is the average squared deviation from the mean

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

- The sum $\sum_{i=1}^n (x_i - \bar{x})^2$ is called the sum of squares

Measures of Spread: Standard Deviation

- Since the variance uses the squared deviation, we usually take its square root called the **standard deviation**

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

- The standard deviation represents (roughly) the average distance of an observation from the mean
- The greater s is the greater the variability in the data is
- We denote the population parameter for the variance and standard deviation using σ for s and σ^2 for s^2

Why divide by $n - 1$?

- We divide by $n - 1$ because we have only $n - 1$ pieces of independent information for s^2
- Since the sum of the deviations must add to zero, then if we know the first $n - 1$ deviations we can always figure out the last one
- Ex.) suppose we have two data points and deviation of the first data point is $x - \bar{x} = -5$
 - Then the deviation of the second data point has to be 5 for the sum of deviations to be zero.

Try it out: computing the variance and standard deviation

- Roll a six-sided die $n = 10$ times and record the number rolled each time
- Data = 1,2,3,3,4,4,4,5,6,6
- Mean = 3.8

Working:

deviations -2.8 -1.8 -0.8 -0.8 0.2 0.2 0.2 1.2 2.2 2.2
(they sum to 0, as they always must)

squared 7.84 3.24 0.64 0.64 0.04 0.04 0.04 1.44 4.84 4.84
sum of squares = 23.6

$$s^2 = 23.6 / (10 - 1) = 2.622$$

$$s = \sqrt{2.622} = 1.619$$

s is back in pips; s^2 is in pips squared.



Which measures do you report?

- A center on its own does not describe a distribution - it needs a spread beside it, and the two come in matched pairs
 - the mean and the standard deviation – both use every observation, so both are pulled by an outlier
 - the median and the IQR – both are positions in the ordered data, so both are resistant
- A measure is **resistant** when a single extreme observation barely moves it
- Roughly symmetric, no outliers: report the mean and the standard deviation
- Skewed, or outliers present: report the median and the IQR
- Do not mix the pair. A mean next to an IQR describes two different things and answers neither question well.

The five number summary

- **Five numbers** are enough to sketch a whole distribution

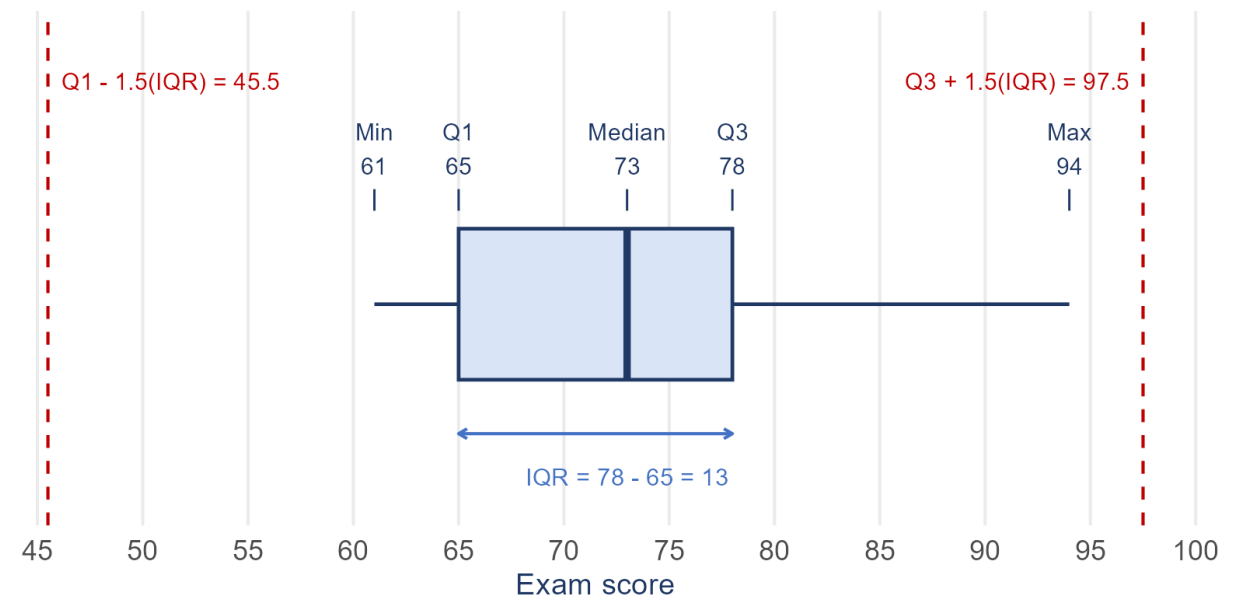
- | Minimum | Q1 | Median | Q3 | Maximum |
|---------|----|--------|----|---------|
|---------|----|--------|----|---------|

- | | | | | | |
|-------------------------|----|----|----|----|----|
| For the 15 exam scores: | 61 | 65 | 73 | 78 | 94 |
|-------------------------|----|----|----|----|----|

- the median gives the center
- Q1 and Q3 bracket the middle half, and their difference is the IQR
- the minimum and maximum give the full extent
- These five numbers are exactly what a boxplot draws

The Boxplot (Box and Whisker Plot): A five number summary

- Pros:
 - good for describing shape and location
 - Can be used to identify outliers
 - Length of whiskers indicates skew
 - Good for comparing two distributions or across categories
- Cons:
 - does not show certain features like mounds, or gaps as well as a histogram



Identifying outliers: the 1.5 x IQR rule

- We need a rule for "extreme" that does not depend on eyeballing the plot
- The **fences** sit an extra one and a half IQRs beyond the box:
 - $Q1 - 1.5(IQR)$ and $Q3 + 1.5(IQR)$
- Any observation outside a fence is an **outlier**, and gets plotted as its own point
- The whiskers then stop at the most extreme observation INSIDE the fences - not at the minimum and maximum
- The fences themselves are never drawn on the plot
- Exam scores: $65 - 1.5(13) = 45.5$ and $78 + 1.5(13) = 97.5$, so nothing is flagged and the whiskers reach 61 and 94
- Being flagged does not make an observation wrong. It makes it worth a second look.

Ex. Construct a Boxplot

- Consider the following data which come from 20 rolls of a six-sided die

• Data =

lower half	middle	upper half
1, 1, 1, 2, 2, 2, 2, 2, 2	2	3, 3, 3, 3, 4, 5, 5, 6, 6, 6

$$Q2 = \text{median} = \frac{2+3}{2} = 2.5$$

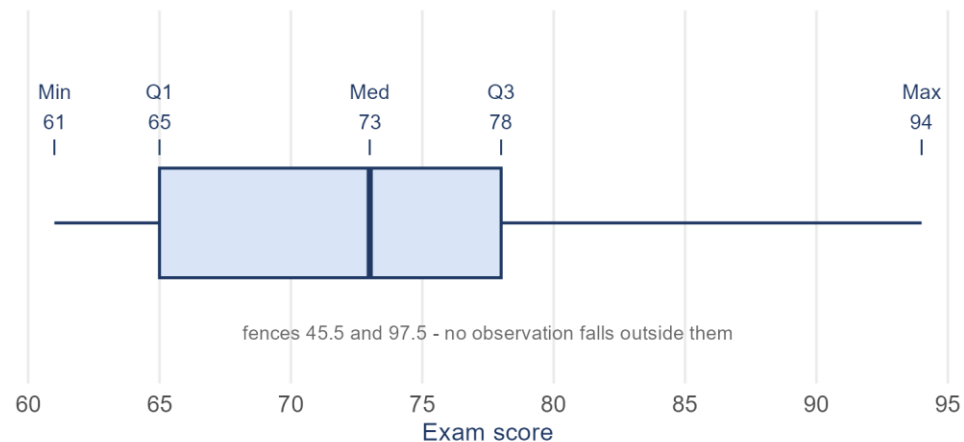
$$Q1 = \text{median lower half} = \frac{2+2}{2} = 2$$

$$Q3 = \text{median upper half} = \frac{4+5}{2} = 4.5$$

Practice: Construct a Boxplot

Construct the boxplot for student exam scores

Exam Scores 61,61,65,65,66,68,69,73,74,75,76,78,79,90,94



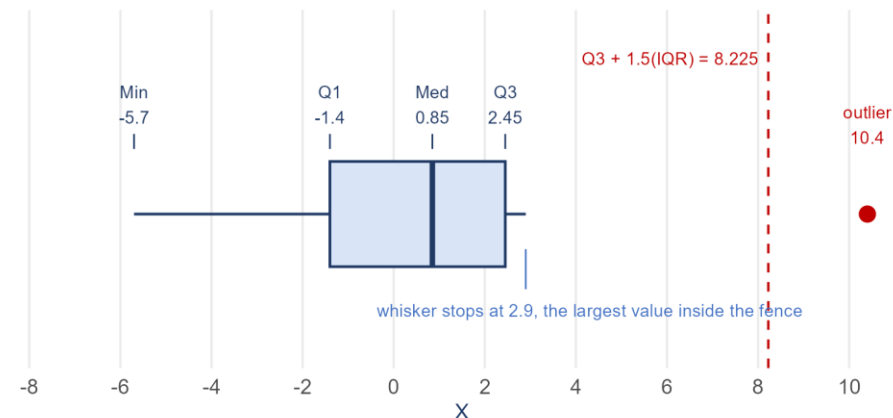
Ex2. Construct a Boxplot

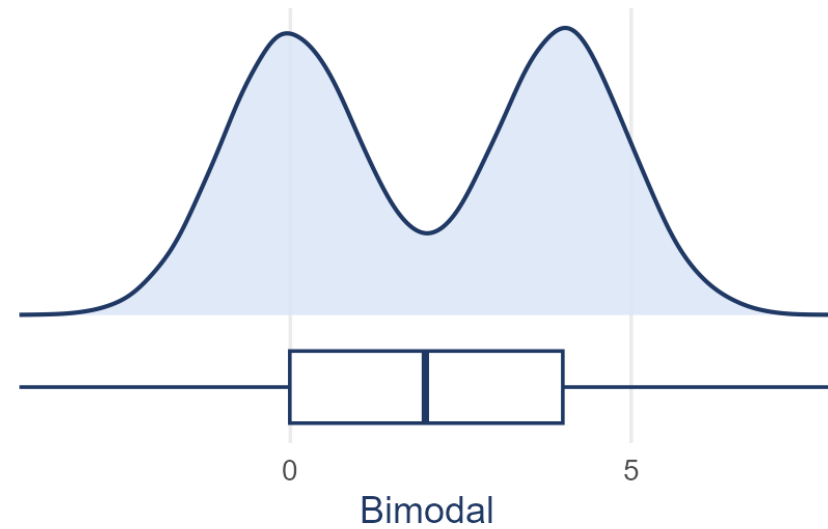
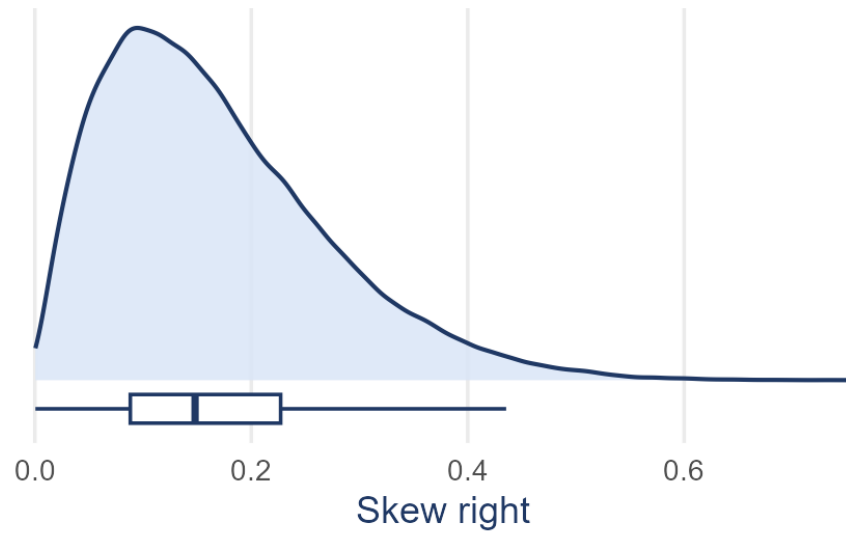
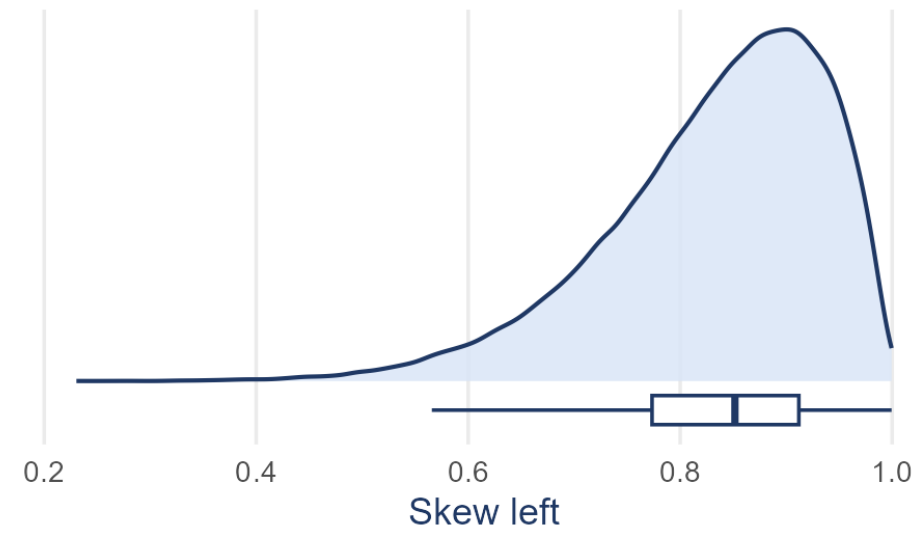
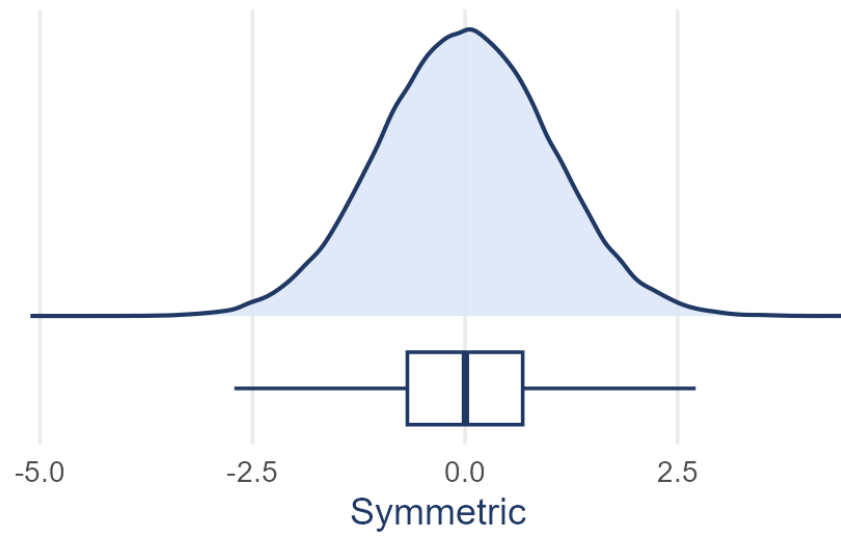
Consider the following 12 observations of a quantitative variable X

$X = \{-5.7, -2.6, -1.5, -1.3, -0.4, 0.2, 1.5, 2.2, 2.3, 2.6, 2.9, 10.4\}$

Compute the 5 number summary and draw a boxplot

Compare the IQR with the range





Whisker lengths show skew. But the bimodal panel has the same boxplot as a symmetric one - a boxplot cannot show peaks or gaps, so read it next to a histogram.